

Mental World Modeling

The shared lexicon for the cluster

The cluster page opens with two challenges and neither of them is a question about cybersecurity. They are conditions on whether the rest of the cluster can be asked at all.

“A long history of work on mental operations never settled on a single canonical definition.”

website, Mental World Modeling, challenge one

“The methods have to recover strategy from sparse, sometimes adversarial traces.”

website, Mental World Modeling, challenge two

The first is a definition problem and the second is a measurement problem. Every project below is one or the other, and the two branches split along exactly that line. Problem-Solving takes the definition problem and pays for it in resolution, holding the vocabulary coarse enough that two people coding one solve will agree. Opponent modeling takes the measurement problem and pays for it by putting structure on the adversary in advance, in a reward function or in a category, so that something is left to recover when the trace is thin.

This file is the vocabulary those two halves share. Six terms appear on more than one page of the cluster, and four of them appear three times in three different wordings. Where a term means two things across two projects, the difference sits inside the entry, in a marked box. That is the entry most worth having, and the census question asks about precisely those.

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1 Three names, and which one is the working unit

The cluster is called Mental World Modeling. The concept its own page says has no consensus is the Mental Model. The unit every project underneath actually works in is the mental operation. Those are three different objects and the difference is not cosmetic, because only one of them is something a coder can be asked to find in a transcript.

Mental Model, mental world model, mental operation

The page raises the definition problem about one term: “lack of consensus on the foundational concept, ‘Mental Model’ ” (website, Mental World Modeling, challenge one). Nothing else in the cluster uses the phrase again.

The cluster name says mental world model. On disk the object of that name is specified: “a directed, multi-relational, weighted hypergraph whose nodes are questions and whose hyperedges are mental operations” (core idea, main.tex, abstract). That is a model of a solver’s world of open questions, not a model inside the solver’s head.

Every project page works in mental operations, and so do all four sections of the framework’s definitions.

The position this file takes: the Mental Model is the inherited term the cluster declines to adopt, the mental world model is the object being built, and the mental operation is the unit of work. The three are kept apart everywhere below.

There is a second collision, in the other branch, and it is a naming collision rather than a sense one. The adversarial half is reached by a link called Opponent (Agent) Modeling, its own page is headed Social World Modeling, the folder on disk is `opponent-agent-modeling`, and the site navigation calls it Strategic Interaction. Four live names for one branch. Its child page has the same problem: the link text is Iterated Strategic Interaction, the file is `social-intelligence.md`, and the heading is Iterated Prisoner’s Dilemma. The entries below use the branch’s own headings, since those are what a reader sees.

2 Active

Terms that have to be defined on the spot, in a talk or in a response, without looking anything up.

2.1 The unit and its scale

mental operation

“a discrete act of thought that transforms one mental state into another”

core idea, sections/on_mental_operations.tex, opening

Definition: As above, held at a stated grain. In the formal version it is a directed, typed, weighted hyperedge $o = (S(o), T(o), r(o), w_t(o))$ over questions, carrying a relation type and a time-indexed confidence that the edge is valid.

Distinguishes it from: An action. An action is the smallest observable unit of solving behaviour, an utterance fragment, a command, a click, a test, and it is observed directly. An operation is

inferred from a window of actions and is never read off one. The gap between them is where all the measurement difficulty sits.

Operational test: Give two coders the same segment and the same label set. If they cannot put the same label on it, what you have is a name, not an operation.

Where it appears: General CS as a twenty-code scheme, twelve object-level and eight regulatory. Capture-the-Flag as six phases with the specific operations living inside them. The core framework as the hyperedge. Small Groups is the one project in the cluster that does not use the term, because its unit is an utterance in a conversation.

grain

“The right grain might depend on the scale we are looking at, the same solve is a handful of steps or a thousand, depending on where you stand.”

website, Mental World Modeling, challenge one

Definition: The temporal and semantic scale at which the operation vocabulary is fixed. Three scales that do not reduce to one another: elementary acts on the order of a fraction of a second, strategies composed of many of them, and regulation over a whole attempt.

Distinguishes it from: The resolution of the recording. Grain is a decision about the label set. A keystroke log is fine-resolution data that can still be coded at Polya scale, and a fifteen-minute transcript is coarse data that can still be coded at sub-second scale and coded badly.

Operational test: Name the duration band the vocabulary is meant to occupy. If one list mixes keystroke-scale acts with whole-attempt heuristics, the grain was never fixed, it was defaulted.

Where it appears: Three uses and all three treat it as an already-named thing: “The grain is coarse” (Capture-the-Flag, two deliberate choices), “the grain problem softens” (Next, operations as learned actions), and “fix the grain deliberately” as the first of four design consequences in the operations essay. Newell’s time scale of human action is the nearest thing the literature has to a shared ordering across the three scales, and it orders operations by duration without saying which operations populate which band.

the control layer

“Knowing an operation is not using it well.”

website, Next, the control layer

“people fail less for lack of heuristics than for lack of regulation, choosing the next move, noticing a line isn’t working, deciding when to abandon it”

same

Definition: The layer that selects among operations, monitors whether a line is working, and decides when to quit. Planning before, monitoring during, evaluating after.

Distinguishes it from: The operations it governs. Regulation is not one more operation of the same kind sitting in the same list. In the CS codebook the two are separated by construction, R1 through R8 above O1 through O12, and the boundary cases are written into the codebook rather than left to the coder: designing the replacement algorithm after a strategy switch is coded as design, not as switching.

Operational test: Does the label describe a change to the problem, or a change to the plan for working on the problem? The second one is regulation.

Where it appears: The cluster page states it as two capacities, “knowing an operation and regulating when to use it are different capacities”. Next names it the under-modeled one. Capture-the-Flag makes monitoring and regulation the fifth of six phases. General CS gives it eight codes. In the cyber setting the two concrete versions are tool selection and knowing when to stop, and both look less like steps inside a solve than like an optimal-stopping problem sitting on top of one.

A citation correction that belongs with this entry, because the cluster relies on it. Flavell 1979 named metacognition and mapped its knowledge component, the person, task and strategy knowledge a solver has about cognition. The regulation triad usually wanted, planning, monitoring, evaluating, is Brown 1987. The two are routinely merged in citations and they are not the same claim.

2.2 The standard the vocabulary is held to

reliability, and what it costs

“a construct nobody can code reliably isn’t a construct”

website, General CS Problem-Solving, the question

Definition: The admission standard for a vocabulary in this cluster. Independent coders agree on where one operation ends and the next begins, or the vocabulary does not enter the analysis.

Distinguishes it from: Validity, which asks whether the label names anything real. Reliability is the cheaper question and it is asked first, on the argument that an unreliable construct cannot be shown valid anyway. The price is resolution: fine schemes are dropped in favour of coarse ones that survive.

Operational test: Three coders, one session, agreement reported before any labels are merged. Agreement computed after merging the labels coders confused is a different number and it is not the one to report.

Where it appears: Stated three times in three vocabularies. “a deliberately coarse, reliability-first account of how people solve” (cluster page, year one). “a construct nobody can code reliably isn’t a construct” (General CS). “Finer metacognitive coding schemes have a history of collapsing under inter-rater disagreement, so the phases stay broad and the specific operations live inside them. That’s a reliability decision, not an omission.” (Capture-the-Flag). The same commitment appears a fourth time in the SIGCSE prep notes as a limit on all of it: whatever set of operations we end up with has to be coarse enough that two people coding the same solve will agree.

disagreement as finding

“Disagreement is treated as the finding rather than the noise: the segments where coders split are the segments where the operation genuinely is ambiguous, and those are worth studying rather than smoothing away.”

website, General CS Problem-Solving, can observers agree on it

Definition: The rule that a split between coders is retained and reported as a location, not resolved by adjudication and then forgotten.

Distinguishes it from: Low reliability, which is a verdict on the scheme. This is a verdict on a segment. A scheme can hit acceptable agreement overall and still have a handful of segments that split every time, and those are the ones that say where the construct is soft.

Operational test: After adjudication, can you still say which segments disagreed and who took which side? If the consensus track overwrote them, the finding was thrown away.

Where it appears: General CS as the running agreement figure that is “deliberately not flattering”. Astrolabe as a design commitment: where coders diverge is treated as a finding, not a defect. The IBL-Group report as the reporting rule that higher discrimination and higher payoff are not the same finding, and that both get reported rather than the more favourable of the two.

2.3 Getting the strategy out of the person

elicitation

“the gold-standard elicitation method, thinking aloud as you go, is exactly what you cannot ask for”

website, Mental World Modeling, challenge two

Definition: The method by which a solver’s strategy is recovered, given that much of what an expert does is tacit and genuinely unavailable to introspection.

Distinguishes it from: Instrumentation. Platform logs are cheap and always available and they tell you behaviour, which is not strategy. Elicitation is whatever gets you from the behaviour to the reason for it.

Operational test: Does the method run during the task or after it? During gives valid contents of working memory and distorts the task. After leaves the task alone and invites invention. Everything else is an attempt to buy some of both.

Where it appears: Capture-the-Flag rules out five methods by name and keeps one. The core framework proposes a different answer to the same problem, validation by convergence across five methods with different blind spots, on the argument that the object is fixed not by one measurement but by the agreement of several.

The six methods, with the verdict the CTF page gives each. This is the densest terminology block in the branch and it is reproduced whole because the verdicts are the content.

Method	Verdict
Concurrent think-aloud	The gold standard, and too intrusive for a timed competition
Retrospective think-aloud	Invites invention
Experience sampling	Interrupts
Cognitive task analysis interviews	Expensive, and depend on having an expert on hand
Platform logs	Cheap, and only tell you behaviour
Stimulated recall	The workable one, anchored to submission logs so the recall has timestamps to hang on

stimulated recall

Definition: Replay the solver their own session as a retrieval cue, after the task, and record what they say about it.

Distinguishes it from: Retrospective think-aloud, which asks for the same report without the cue. The cue is the whole difference, and it is what makes the report a recall rather than a reconstruction.

Operational test: Is there a timestamped record to anchor the replay to? Without one there is nothing to hang the recall on and the method degrades into the retrospective case.

Where it appears: The cluster page names it as the branch’s answer to challenge two. Capture-the-Flag anchors it to submission logs. General CS reaches the same problem from the other side, with transcripts recorded in the browser and transcribed, then coded.

2.4 Reading the other agent

type, and typing another agent

“recovering a hidden type from behaviour, while the other party adapts”

website, Next, one account of typing another agent

Definition: A compressed category assigned to another agent from what that agent has been observed to do, standing in for a full model of them.

Distinguishes it from: A strategy. A strategy is what the other agent does. A type is what the modeller carries instead of the strategy, because carrying the strategy is too expensive. The compression is the point, not an approximation forced on you.

Operational test: Is the category assigned against a fixed threshold or against the rest of the population? Relative assignment means the same behaviour lands in different categories depending on who else is in the room, and that changes what the type is a claim about.

Where it appears: Three formulations of one thing. “how agents read each other into kinds across repeated interaction” (cluster page, year one). “It compresses them into a small number of types, cooperative, conditional, exploitative” (Iterated Prisoner’s Dilemma, categorization). “one account of how an agent types another” (Next). The adversarial version carries none of this vocabulary and calls the same object a behavioural fingerprint.

Two senses of type, and the cluster says so itself

On the cooperative page a type is an empirical category the focal agent builds from a running estimate of how often each partner cooperates, assigned relative to the group mean.

On the Next page a type is hidden and is recovered from behaviour while the other party adapts, which is the type of a Bayesian game: a property the other agent already has and the modeller does not observe.

The two are not obviously the same object. One is constructed by the observer and would not survive a change in the reference group. The other is a fact about the observed agent. The cluster’s own open problem asks for them to be folded into a single account, which is the right request and is not the same as asserting they already are one. Left open here.

realistic opponent

“so that the defender it trains faces a realistic opponent instead of an idealized one”

website, Mental World Modeling, year one

Definition: An adversary model that departs from optimal play in structured, repeatable ways, rather than an expected-utility maximiser.

Distinguishes it from: A hand-written rule set, which also departs from optimal play but does not generalise past the cases that were written down. The biased-attacker line puts the departure in the reward function instead, so it generalises by construction.

Operational test: Change the environment and see whether the departure survives. A bias in the reward moves with the agent. A rule stays where it was written.

Where it appears: The cluster page as the reason the adversarial branch exists. Biased Attacker as the mechanism, prospect theory’s value function carried into the POMDP with three parameters. Challenging Attacker as the two scripted routes, B-line and Meander, which are the idealised case the rest of the branch is trying to get past.

behavioural fingerprint

“If those deviations are systematic and separable, they are behavioural fingerprints: you could tell one kind of adversary from another by how they move”

website, Biased Attacker, why it matters

Definition: A pattern in an agent’s trajectory that identifies which kind of agent produced it, given that the network, the action space and the objective were held fixed.

Distinguishes it from: A signature in the detection sense, which identifies an attack. A fingerprint identifies an attacker, and it survives the attacker changing which attack they run.

Operational test: Two conditions, both required, and the sentence states them. Systematic, so the deviation recurs. Separable, so two kinds do not produce the same pattern.

Where it appears: Biased Attacker as the payoff of the whole design. The IBL-Group work reaches the same construct from the cooperative side and calls it a five-feature behavioural profile over a partner. This is the term `dict_opponent_modeling.tex` has to carry across the two vocabularies.

2.5 The object being built

question graph

“the incremental construction of a directed, multi-relational, weighted hypergraph whose nodes are questions and whose hyperedges are mental operations, reconstructed from a solver’s action trace”

core idea, main.tex, abstract

Definition: As above. A problem is a pair of initial questions and a goal predicate, a solving state is a sub-hypergraph, and a solution is a minimal set of operations reaching a goal question, which is a hyperpath because every source of an applied operation is required.

Distinguishes it from: A concept map, which has concepts as nodes. Here concepts are carried by the questions and the concept graph is a projection of the question graph onto concept space, not the primary object. Also from a sequence: a session is not a list, and the branching and the loops are the part worth having.

Operational test: Can an edge have more than one source question, with all of them required? If not, it is a graph and the hypergraph machinery is unused.

Where it appears: The core framework as the definitions. General CS as the solve drawn as it happens, and as the qcards and origin edges in Astrolabe. Capture-the-Flag as the difficulty construction. Its maturity is stated in the SIGCSE notes and is worth carrying: definitions drafted, zero data through the framework itself.

difficulty

“The difficulty of a problem, for a solver θ , is a vector functional $D_\theta(P)$ rather than a single number, because its three sources are not commensurable.”

core idea, main.tex, definitions

Definition: Three components. Solution size, the length of the optimal solution. Concept load, the number of concepts held and related at once. Search burden, the effort of finding the operations rather than applying them. When no solution exists difficulty is maximal rather than undefined, since certifying that a closed problem has no solution is itself a co-NP task.

Distinguishes it from: An item difficulty parameter, which is one number fitted to a population. This one is relative to a solver and is a vector, and the two facts are connected: difficulty and ability are two sides of one relation, so a difficult problem is one only a capable solver can crack.

Operational test: Can two problems be incomparable? Under a vector they can, and that is the intended behaviour. If your construction always returns an ordering, you have collapsed the three sources into one and should say with what weights.

Where it appears: The core framework as the definition. Capture-the-Flag as its own index question, what makes a CTF challenge difficult. Note the sense split against the deployment cluster, where difficulty is an ordinal per-dimension priority in a use-case table and not a vector at all.

3 Working

Recognized precisely, lookup acceptable. Definition and discrimination only.

action

Definition: The smallest observable unit of solving behaviour, an utterance fragment, a command, a click, or a test, taken as a primitive and observed directly.

Distinguishes it from: The operation inferred from it. Action mining inherits the ambiguity of where one unit of behaviour ends and the next begins, which is not fixed across utterances, commands, and tests.

concept

Definition: An atomic semantic unit that a question concerns, extracted from a window of actions.

Distinguishes it from: A fixed dictionary entry retrieved unchanged. The framework treats drift as a feature rather than a defect, on the observation that human solvers drift constantly and much of the work of understanding happens in that drift. Elsewhere in the same folder the term is settled differently, as a coordination handle rather than a claim about representational kind, and that difference is deliberate.

relation type

Definition: The label every operation carries, from a vocabulary of four: compositional, the part-whole relation; causal, resolving the sources enables the target; co-occurrence, undirected, no claim of order or cause; dependency, the target cannot be resolved until its sources are.

Distinguishes it from: Chronological order, which is what a trace gives you for free and which none of the four is. Dependency is the one that induces an acyclic graph, the prerequisite order among questions.

criticality

Definition: The value of information of a question on the dependency graph.

Distinguishes it from: The causal relation, and the framework says so explicitly: a question can be a prerequisite without being a cause. Also from the hinge, which is the same idea reached from the data rather than from the definition, the step that turned out to decide the solve.

trace

Definition: The observed sequence of actions in one session, from which the graph is reconstructed.

Distinguishes it from: The graph. The structure inference problem is exactly the gap between them, and the framework's second proposition bounds it: the operation labeling is recoverable from a trace only up to a tolerance, since distinct labelings can fit the same trace.

inter-rater agreement

Definition: Stage-level percent agreement and Cohen's kappa between coders labelling the same session.

Distinguishes it from: Consensus, which is what you build after measuring agreement. Reporting the consensus track without the agreement figure hides the number the scheme is being judged on.

categorization and contrast

Definition: Categorization compresses many partners into a small number of types. Contrast makes the assignment relative: a partner is classified against the group mean, so identical evidence yields a different judgement depending on who else is in the room.

Distinguishes it from: An absolute threshold rule, which is what most opponent classifiers use. The reconstruction work found the printed contrast formula is a similarity measure while the

mechanism it implements needs a dissimilarity measure, and that discrepancy is recorded rather than repaired silently.

episode

Definition: One bounded stretch of solving, segmented from the trace, which a question and its operations belong to.

Distinguishes it from: The session, which is the whole recording. One session contains many episodes and the episode boundaries are inferred, which is where coders most often split.

4 Peripheral

Known roughly, with a pointer to where it is used.

options, and library learning

Two existing answers to the cluster's first open problem: whether an agent could discover its own operations rather than be handed them. Hierarchical reinforcement learning learns options over primitive actions. Program-synthesis systems build a library of reusable subroutines and reuse it. Both are named on the Next page as the shape the answer might take.

optimal stopping

The formal home the Next page proposes for the control layer in the cyber setting. Tool selection and knowing when to stop are treated as a problem sitting on top of the solve, not as steps inside it.

prospect theory value function

The mechanism the Biased Attacker project uses to make bias generalise. Three parameters, curvature on gains and losses for risk sensitivity, and a loss-aversion multiplier. Detail in `mwm_biased_attacker.tex`.

instance-based learning

The cognitive architecture the whole opponent-modeling branch is built on, and the reason the categorization and contrast mechanisms take the form they do. Detail in `mwm_iterated_prisoners_dilemma.tex`.

Polya, Schoenfeld, Newell and Simon

The four inherited accounts the operation vocabulary is assembled from. Polya supplies an episode scaffold and not a chronology. Schoenfeld supplies the split between resources, heuristics, control and beliefs, which is why possessing a strategy is treated as a different capacity from selecting it. Newell and Simon supply the operator over problem states, which is what the hyperedge is a generalisation of.

5 The three open problems

The cluster's Next page states them, in its own headings, and they are the reason this lexicon is arranged the way it is.

Operations as learned actions, not a fixed taxonomy. Every account in the cluster takes its operations as given, named in advance and then looked for in the data. If they were learned and composable instead, the grain problem softens, because the agent finds the grain that pays, and the model stops depending on a taxonomy no one has agreed on.

The control layer. A distinct problem from the operations it governs, and the under-modeled one.

One account of typing another agent. The cooperative side reads a partner into a kind, the adversarial side models an attacker’s built-in bias, and the page’s claim is that these are the same inference done twice with different vocabulary.

The third is the one this file can act on now, because both vocabularies are already written down and they can be laid against each other without new data. That is what `dict_opponent_modeling.tex` does. The first two need the data the cluster does not yet have.

6 What crosses out of this cluster

Terms used here that another cluster also uses, in a sense that is not the same. Each stays in both files and each entry says how the other cluster uses it.

Term	Here	In Cyber World Modeling
world model	A hypergraph of a solver’s questions and operations	A learned per-device latent dynamics model with a controllable, exogenous and opponent factoring
opponent model	A category assignment over a partner’s behavioural profile	A per-device embedding inferred from defender-observable signals
operation	A discrete act of thought, or a typed hyperedge	Not used; the nearest term is the defender action
difficulty	A three-component vector over solution size, concept load and search burden	Not used; the nearest term is sample efficiency
transfer	Whether taught heuristics carry to a new problem	Policy transfer across environments

The first row is the one to watch. Two folders on disk are both called world modeling and they share almost no vocabulary. Whether the parallel is a claim or a naming convenience is not settled anywhere in either folder, and this file does not settle it either. What it does is keep the two definitions on one page so the question can be asked with both in view.

7 Sources

- Every quoted sentence in the Active tier is from the cluster page, the Next page, one of the six project pages, or the core framework on disk. The locator on each entry says which.

- The formal definitions of question, operation, relation type, criticality, solution, cost and difficulty are from `core idea/main.tex`, section Definitions. The four propositions and the identifiability bound are from the section that follows it.
- The account of grain, the three scales, and the Flavell against Brown correction are from `core idea/sections/on_mental_operations.tex`.
- The reliability commitment is quoted from three pages and appears a fourth time in the SIGCSE prep notes.
- The three tiers and the four fields are from the Tech Fluency page. The reason a term whose two uses disagree belongs inside its own entry is in `_side/_lexicons/lexicon_method.tex`.
- Per-project entries live in six files, one per project: `mwm_small_groups`, `mwm_general_cs`, `mwm_capture_the_flag`, `mwm_iterated_prisoners_dilemma`, `mwm_challenging_attacker`, and `mwm_biased_attacker`. Each points back here for the terms above.